Script-aware BM25 retrieval for Urdu and Roman Urdu news search

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Abstract

Urdu news users often type native Perso-Arabic script, informal Roman Urdu, or a mixture of both, while a single native-script index will not match Latin-script queries even when the article exists. This paper reports a frozen lexical retriever, M0, on 111,860 Urdu news articles. A deterministic Unicode count of Perso-Arabic versus ASCII letters selects one of two Okapi BM25 indexes: Urdu, mixed, and other queries search native-script BM25; Roman queries search a second index whose documents were romanized with a closed character table and a 198-key reverse dictionary (Method D). The detector is not a learned classifier. Queries are not rewritten. BM25 uses $k_1 = 1.5$ and $b = 0.75$; lists are retrieved to depth 50 and scored at a Top-5 cutoff.

On a development/validation known-item pool of title-derived queries, ExactSource Hit@5 was 68/78 (87.18%; Clopper–Pearson 95% interval 77.68–93.68%). On a newly sealed known-item sample the same frozen system achieved 27/40 (67.50%; 50.87–81.43%). On a separate sealed naturalistic set with no gold article, the first author (who also wrote the U queries) judged the Top-5 useful (at least one relevant or partially relevant document) for 23/40 queries (official A1 Success@5 57.50%; 40.89–72.96%). An independent second annotation (A2) is a reliability check only and does not replace 23/40. Query-side Roman expansions did not raise the 68/78 development score, so the freeze was not replaced. In the naturalistic sample, Urdu-script queries succeeded in 17/18 cases and Roman queries in 6/18; all four mixed queries failed.

These three percentages answer different questions and must not be averaged. On the freeze pool, 68 of 78 title-derived known items reached the Top-5. Ordinary Roman Urdu remains the main limitation. This is a hashed empirical evaluation of script-aware BM25, not a learned or online index-selection policy.

Introduction

Urdu is written in a cursive Perso-Arabic script, but that is not how everyone searches. A large share of users type Roman Urdu: Urdu words in Latin letters, with spelling that is informal and inconsistent [1–3]. A news collection indexed only in native script will not match those queries even when the article is sitting in the index. That mismatch is a retrieval failure, not a missing document.

Urdu remains thinner in shared information retrieval (IR) evaluation than English [4]. Collections exist. CURE provides an Urdu ranking resource [5]. An Urdu MS MARCO baseline reports MRR@10 of 0.247 in the abstract (0.248 in the results table) for a fine-tuned Urdu mT5-mMARCO configuration, under a passage-ranking protocol that is not the one used here [6]. Those resources do not freeze a script-conditional lexical system and then keep known-item recovery separate from human usefulness. Roman Urdu research is still mostly classification—sentiment, offensive language—rather than news search [1, 2]. Multilingual dense retrievers can lose effectiveness on languages that sit outside the pretraining head [7], even when scaled multilingual encoders help related tasks such as Urdu natural language inference [8]. Users of lower-resource varieties are also often pushed toward high-resource query forms [9]. Typing Roman Urdu into an Urdu-script index is a concrete instance of that pressure.

The ULTRA framework of Bashir, Qaiser, and Hussain supplies a dual-embedding Urdu news architecture [10]. Its original switch is a static character-length threshold. Length is easy to implement and easy to fool: a short “why” question may need the article body, and a long factoid may be answered by a headline. Length also does not decide which *script index* to open. English work on query routing chooses among sparse and dense retrievers [11] or varies retrieval depth in retrieval-augmented generation [12]. Those systems were not designed for Roman Urdu, and they are not what is evaluated here.

This paper asks a narrower empirical question that can be answered with a freeze and a protocol. Specify a lexical retriever. Hash the corpus and the Roman dictionary. Then measure three things that are easy to mix and should not be mixed. First, how often does the system recover a designated source article from title-derived queries on the development/validation pool used to select the Roman path? Second, does that known-item score hold on a newly sealed sample of title-like queries written after the freeze? Third, how often is the Top-5 useful when there is no gold article and a person judges the hits?

Official retrieval here is Okapi BM25 [13] with a deterministic Unicode script detector that selects one of two lexical indexes. We call that frozen configuration M0. In this paper, *script-aware index selection* means only that choice: URDU, MIXED, and OTHER queries open Urdu BM25; ROMAN queries open Method D. It is not online adaptation, not query rewriting, and not a learned SHORT/LONG classifier. Earlier work in the same project trained a SHORT/LONG support vector machine and a MiniLM dual-index using a Sentence-BERT encoder [14]. Those experiments are historical development, not the official retriever. We mention them only where they were actual development comparators, so that older classification or dense-index numbers are not mistaken for M0.

The contribution is empirical: a hashed freeze of that script-aware BM25 pipeline on one Urdu news collection, evaluations that keep known-item recovery separate from human usefulness, and a documented limitation on ordinary Roman Urdu. We do not claim a new learned routing algorithm, state of the art against CURE or Urdu MS MARCO, or a general solution to Roman Urdu search.

Research questions

- **RQ1.** Can a script-aware lexical pipeline (Urdu BM25 plus Method D) recover known news articles from title-derived queries on a development/validation pool?
- **RQ2.** Does that known-item score transfer to a newly sealed known-item sample written independently of the freeze set?
- **RQ3.** How often is frozen M0 useful—at least one relevant or partially relevant document in the Top-5—on new naturalistic queries with no gold article?
- **RQ4.** Do query-side Roman expansions improve development/validation ExactSource Hit@5 enough to replace M0?

Materials and methods

Research design

The study is an offline IR evaluation of one frozen lexical system. Method selection used a pre-registered development split. The official unseen tests were sealed before retrieval. No test query was used to choose BM25 parameters, the script-to-index rule, the Roman method, or the dictionary. After the freeze, later query-side expansions were accepted only if they improved the development known-item score; they did not, and M0 was left in place.

Two tasks are reported. Known-item search asks whether a pre-assigned source document appears in the Top-5. Naturalistic search has no source identifier; a human labels the retrieved Top-5. Those tasks are not averaged.

Official results are the development/validation known-item pool ($n = 78$), sealed K001–K040, and sealed U001–U040 with Annotator-1 labels. Phase 11 (M1–M4), H001–H040, and dense-index pilots are diagnostic or historical. After those official scores were frozen, later exploratory development on a separate git branch is outside this paper (Limitations).

Evaluation hierarchy

Four layers must not be mixed.

- **A. Freeze-pool known-item** ($n = 78$). Metric: ExactSource Hit@5 (secondary nDCG@5, MRR). Used to select Method D and freeze M0.
- **B. Phase 12 K** (K001–K040). Same known-item metric on a sealed sample. Not human usefulness.
- **C. Phase 12 U** (U001–U040). Official Annotator-1 human Success@5 (secondary P@5, nDCG@5, MRR). No `source_doc_id`.
- **D. A2.** Independent labels on the same 200 U Top-5 documents. Reliability only. Does not replace C.

H001–H040 are a historical diagnostic. They are not A–D.

Corpus

The official collection is `data/clean_articles.csv`: 111,860 news articles (540,050,203 bytes). The local precursor is `data/urdu_news.csv` (111,861 CSV records). That precursor is schema- and size-consistent with Version 1 of the Kaggle listing “Urdu News Dataset” compiled by Shahane (file `urdu-news-dataset-1M.csv`; displayed size 276.79 MB) [15]. The Kaggle page cites Hussain, Mughal, Ali, Hassan, and Daudpota, “Urdu News Dataset 1M”, Mendeley Data, V3, doi:10.17632/834vsxnb99.3 [16]. A SHA-256 comparison of a fresh Kaggle or Mendeley download to the local precursor was not completed. The compilation title uses “1M”; the Kaggle Version 1 explorer and the local precursor contain 111,861 records, not one million. Whether that Kaggle file is the complete Mendeley dump was not verified.

The last precursor record is truncated (three CSV fields instead of eight; `Index` 111860; no trailing newline). `pandas.DataFrame.dropna()` dropped that incomplete row, leaving 111,860 articles. Headline and body were then concatenated:

```
combined_text = Headline + ' ' + News Text
```

There was no Unicode NFKC folding, no ye/he/kaf canonicalization, no diacritic stripping, no stemming, no stopword removal, and no deduplication at indexing. HTML, punctuation, English tokens, and near-duplicate articles were left as in the source dump. Document identifiers equal the corpus row index after that cleaning step. Article categories in the dump are Sports, Business & Economics, Entertainment, and Science & Technology.

Corpus SHA-256: `8992a6acca3459eea17a7d7356dd490445daa00b958eab765713853c97a9f231` (540,050,203 bytes). The Roman dictionary is `models/roman_urdu_dict_expanded.json` (198 keys;

SHA-256 `30c3f61a64ec641abbb3acdbc7a8bcaf197f0238f1bf9e76c2c7ce8e590f86a3`). Preprocessing cells are in `archive/historical_experiments/notebooks/01_preprocessing.ipynb`. Reconstruction notes are in `experiments/publication_audit/`.

Official frozen system (M0)

Fig 1 summarizes M0. Internally the system retrieves 50 documents; the official cutoff is 5. Queries are not rewritten on the official path.

Fig 1. Official frozen retriever M0. A deterministic Unicode script detector sends URDU, MIXED, and OTHER queries to Urdu BM25 and ROMAN queries to Method D romanized-document BM25. Queries are not rewritten. This is not an SVM SHORT/LONG classifier and not a MiniLM dual-index pipeline.

The detector counts characters in the Arabic/Urdu block U+0600–U+06FF and ASCII letters A–Z/a–z. If both counts are zero the label is OTHER. If both are positive the label is MIXED. If only the Urdu count is positive the label is URDU. Otherwise the label is ROMAN. URDU, MIXED, and OTHER search Urdu BM25 over `combined_text`. ROMAN searches Method D BM25 over romanized documents. The detector is not a classifier. On the Phase 2 development/validation pool it agreed with the oracle script tags on 78/78 queries. The nine MIXED items in that pool were mixed by construction (an English template suffix), not detector errors.

Tokenization is the same on both indexes: the regular expression `[\u0600-\u06FF]+|[A-Za-z0-9]+`, lowercase, no stemming, no stopword list. BM25 is Okapi with frozen $k_1 = 1.5$ and $b = 0.75$. Inverse document frequency for term t with document frequency n in a collection of N documents is

$$\mathrm{idf}(t) = \log\left(\frac{N - n + 0.5}{n + 0.5} + 1\right). \tag{1}$$

The implementation is the custom `BM25` class in `experiments/phase5_roman_urdu/run_phase5.py`. The Urdu index has 371,368 terms and mean document length 279.24 tokens. The Method D index has 358,497 terms and mean document length 279.19 tokens.

Method D

Method D keeps the typed Roman query and indexes romanized documents. Every article in the 111,860-document collection is romanized; the index is not restricted to evaluation source documents. After tokenization, each token is handled as follows. If it contains Urdu letters and that exact token is a value in the 198-key dictionary, emit the first reverse-mapped Latin key (`setdefault`; first key wins). Otherwise emit a character-table romanization (`_CHAR_ROMAN` from the Phase 2 pipeline). Tokens that are already Latin or alphanumeric are lowercased and kept. The character table maps, among other pairs, madda alif to `aa` and che to `ch`, and collapses several Urdu letters onto the same Latin letter (for example te and tte both map to `t`). That collision is intentional and lossy.

Method D is the document-side counterpart of Phase 2 `title_roman` generation. It is not a mapping fitted to evaluation query strings.

Roman-method selection (Phase 5)

Four Roman strategies were pre-registered and compared on development Roman known-item queries only ($n = 13$), before internal-validation scores were interpreted. Selection used ExactSource Hit@5, then nDCG@5, then lower mean query latency. BM25 hyperparameters were not retuned. Method comparison counts are in S3 Table.

Method A sends the raw Roman query to the Urdu-script BM25 index. Method B transliterates with the existing 198-key dictionary and then searches that same Urdu index; unmapped tokens stay Latin. Method C applies a closed spelling-variant table, dictionary lookup, and a greedy inverse of the Phase 2 character table, then searches Urdu BM25. Method D is the romanized-document index described above. A fifth analysis, Method E, reported the union of C and D; it was not a selectable system, and reciprocal rank fusion was not built [17].

The selected method was frozen in `artifacts/selected_method.json` before internal validation ($n = 10$ Roman queries) was read. Mixed queries under the deployable rule go to Urdu BM25. Urdu queries always use Urdu BM25. Method D was selected on $n = 13$ development Roman queries.

Development/validation known-item pool

Phase 2 built 260 title-derived known-item queries from corpus articles that were not used in earlier trap sets. Each query has one `source_doc_id`. Templates include short title, romanized title, why/how/effects, lead excerpt, and mixed Urdu+English. Roman strings in this pool are `title_roman`: dictionary reverse lookup plus the character table applied to headlines. They are not chat-style Roman Urdu. The split (seed 42) is train $n = 182$, development $n = 39$, internal validation $n = 39$. Official M0 development/validation reporting uses development plus internal validation ($n = 78$). The train split was not used to select Method D; it was later used only as a diagnostic pool for query-side expansions.

Query-side expansions (Phase 11)

After M0 was frozen, four query-side Roman transforms were scored on the same $n = 78$ pool. The hard gate was ExactSource Hit@5 not below 68/78. A train-Roman diagnostic ($n = 64$) was used only among candidates that passed the gate. H001–H040 and the later sealed sets were not loaded.

M0 applies no query transform. M1 appends a closed alias list while keeping original tokens. M2 adds four further aliases. M3 applies M1 then drops a nine-token query stoplist. M4 combines M1, M2, and the stoplist. The 198-key dictionary file was not edited. Terms such as `diesel` and `iphone` were explicitly forbidden as new keys. The official Phase 12 run did not apply M1–M4.

Sealed evaluation (Phase 12)

K001–K040 (known-item) and U001–U040 (naturalistic) were written and checksum-sealed before retrieval (seed 120260827). Query-file SHA-256 values are `124e452693f98baedf510618240c154df68d56b6b7a37ed085a6512c13d13ff6` (K) and `684fd1e19eddb717f5897d869ef0ca0ed586316c5a7e1d2d23006e0748fc53b9` (U). One retrieval pass of frozen M0 produced Top-50 lists. Queries were not edited after ranks were seen. M1–M4 were not applied.

K queries are shortened or lightly paraphrased headlines from articles that were not Phase 2 QTRN sources (260 source identifiers excluded; eligible population 111,574 headlines of length at least 12 after stripping). The creator could see the source article in order to assign `source_doc_id` and did not run BM25 while writing. The 12 Roman K queries are ordinary Roman Urdu of the selected headline, not `title_roman` character-table strings. No mixed-script K query occurred naturally; mixed script was not forced.

U queries follow pre-registered naturalistic quotas: 18 Urdu, 18 Roman, 4 mixed; 12 short (≤ 5 whitespace tokens), 16 medium (6–12), 12 long (13+); 14 factoid, 14 explanatory, 12 named-entity. Four temporal factoids use *aaj* / `aaj` / `mojooda` forms. U has no `source_doc_id`. Queries were not copied from H001–H040. After retrieval, detector counts were 18 URDU, 18 ROMAN, and 4 MIXED; retrieval-path counts were 22 Urdu BM25 (URDU plus MIXED) and 18 Method D. Thirty-nine queries returned 50 hits; U006 returned 28. No query returned fewer than five hits.

Human relevance protocol (U only)

After the sealed Top-5 dump existed, the first author labeled 200 documents (40 queries $\times$ 5) from headline and snippet only. The same author had written the sealed U queries under the Phase 12 protocol. Labels follow a five-way rubric: A, relevant (the reader could stop); B, partially relevant (same event or occasion, incomplete answer); C, topically related (same topic or entity, not the asked need); D, not relevant; E, ambiguous if A–D cannot be decided. The annotator preferred B over A unless the need was clearly satisfied, and preferred C over B unless the article helped answer the asked need. Temporal queries were judged as archive type-of-fact for a dated occasion in the article, not against the annotator's calendar day. Recurring wires with no date in the query (gold price, budget date, eclipse, stock close) could be A even if dates differed. Named-entity lookups were A if the article's main subject was that person. The original evaluation labels (A1) were retained as the official evaluation labels. An independent second annotation (A2) was subsequently conducted by Areena Rahman on the same 200 query-document judgments as a reliability check; A2 was not used to replace or recompute the reported A1 results (S2 File; per-query labels in S4 Table). Official Success@5 remains Annotator 1 (23/40). ExactSource Hit@5 was not computed on U. Label definitions are in S1 Text.

Ethics

Relevance labels for U001–U040 were assigned by annotators to retrieved public news headlines and snippets. No search-user logs, interviews, or other identifiable participant records were collected for this evaluation. No institutional ethics-board determination is recorded in the project repository.

Metrics

For a known-item query with source identifier s, ExactSource Hit@k is 1 if and only if s appears in the retrieved top k. The official cutoff is $k = 5$. Secondary K cutoffs are 1, 10, and 50. Known-item nDCG@5 on the development pool uses gain $1/\log_2(\text{rank} + 1)$ when the source is in the Top-5, else 0. Known-item P@5 equals 0.2 times Hit@5 because there is a single relevant document.

For naturalistic queries, Success@5 is 1 if and only if at least one Top-5 document is A or B. Conservative P@5 is the mean of (count of A labels)/5. nDCG@5 uses gains A = 3, B = 2, C = 1, D = E = 0. MRR is the reciprocal rank of the first A or B, or 0 if none. nDCG@5 is secondary: a Top-5 of only topical C documents can score nDCG@5 = 1 with no A or B.

Clopper–Pearson 95% intervals were computed from the reported binomial counts using SciPy's `binomtest` (SciPy 1.16.3). They were not part of method selection. No hypothesis test comparing systems on K or U was pre-registered; we do not report p-values for those splits.

Rejected alternatives

Development comparators on the same $n = 78$ pool include headline MiniLM, a truncated full-article Chroma index, a 96/32 chunk approximate nearest-neighbour index, and Urdu-only BM25. A pre-registered long-context `intfloat/multilingual-e5-small` index was not built: a CPU prototype ran at 2.3 documents/s (about 13.5 hours extrapolated for 111,860 articles) and failed a 4-hour gate. The encoder was not swapped after that failure. Fusion of headline and script-aware lists was not deployed: an oracle union added three known-item hits, which was judged insufficient. A reranker was not built. Historical SHORT/LONG SVM index selection and MiniLM dual-index graded P@5 on H001–H040 are a separate development study. They are not official M0 metrics.

Diagnostic set H001–H040

H001–H040 are historical trap strings with no source identifier. ExactSource Hit@5 is undefined on that set. A later human pass (Phase 10C) labeled 196 retrieved rows (38 queries × 5, plus H027 × 5 and H036 × 1; H036 returned a single document). Success@5 was 25/40; conservative P@5 was 0.1250; 10/40 lists were all D. On that sample, Success@5 was 14/20 for Urdu-script queries and 11/20 for Roman; eight of the ten all-D lists were Roman. Labels were assigned from headline plus a 500-character snippet. Query text, rank-1, and labels are now known. The set is treated as a historical diagnostic and is not the official unseen usefulness result. It is not combined with U.

Software environment and reproducibility

M0 retrieval uses Python 3.13.9 (Anaconda, 64-bit Windows). Dense-index pilots in the same project ran on CPU (12 cores reported for the e5-small prototype). Method D index construction on the full corpus took 100.6 s (66.3 s tokenize and romanize, 34.3 s BM25 postings) and occupied 116.1 MB of in-memory postings. Entry points were not relocated: detector and BM25 in `experiments/phase5_roman_urdu/run_phase5.py`, character table in `experiments/phase2_oracle/run_phase2_pipeline.py`, sealed runner in `experiments/phase12_new_unseen_evaluation/run_phase12.py`. The freeze manifest is `experiments/phase8_final_freeze/FINAL_SYSTEM_MANIFEST.json` (S1 File). A SHA-256 check for a local copy of the frozen corpus is `experiments/publication_audit/verify_corpus_hash.py`. Reproduction steps, environment pins, and data-access limits are in the repository file `REPRODUCE.md` on branch `publication/plos-one-final`. Code, the Roman dictionary, sealed queries, qrels, and reconstruction notes are in git. Full retrieval reproduction requires a local copy of the third-party article-text corpus; a git clone does not contain that CSV. This manuscript does not claim that a clean-clone full-corpus rerun was completed for publication.

Data availability

The retrieval collection is a locally processed copy of a third-party Urdu news compilation titled Urdu News Dataset 1M [16]. The local precursor `data/urdu_news.csv` (111,861 records) is schema- and size-consistent with Shahane, Urdu News Dataset, Kaggle Version 1, file `urdu-news-dataset-1M.csv` [15]. That Kaggle listing cites Hussain, Mughal, Ali, Hassan, and Daudpota, Mendeley Data, V3, doi:10.17632/834vsxnb99.3 [16]. A SHA-256 identity check between a fresh provider download and the local precursor was not completed. The compilation title uses “1M”; the file used here has 111,861 precursor records and 111,860 frozen articles.

The frozen file used for all official M0 scores is `data/clean_articles.csv` (111,860 articles; 540,050,203 bytes; SHA-256 `8992a6acca3459eea17a7d7356dd490445daa00b958eab765713853c97a9f231`). Preprocessing dropped one truncated precursor record with `dropna()` and concatenated headline and body as described in the Corpus subsection. The authors do not redistribute the full article-text CSV in the project GitHub repository or in Supporting Information. Redistribution permission for the underlying news article text has not been independently verified. Researchers should obtain the source dataset from Kaggle and/or Mendeley under those providers’ terms, then verify any reconstructed file against that SHA-256. Reconstruction notes without article text are in S3 File. The Roman Urdu dictionary (`models/roman_urdu_dict_expanded.json`; 198 keys), M0 code, sealed query files, human qrels, and reconstruction notes are at `https://github.com/HashimAbbasii/Dynammic-Query-Routing-Urdu-ILS` (branch `publication/plos-one-final`; `REPRODUCE.md`). A git clone of the default branch does not by itself select this submission snapshot. A git clone does not contain the article-text corpus. Official metrics were copied from sealed Phase 8–12 reports and were not recomputed for this manuscript.

Results

Table 1 reports the three official headlines. The rows are different tasks. They are not a single accuracy trajectory.

Development/validation known-item

ExactSource Hit@5 on development plus internal validation was 68/78. Secondary known-item metrics on that pool were nDCG@5 = 0.8107 and MRR = 0.797. An independent rebuild in a later phase reproduced Hit@5 = 0.8718.

Table 2 and Fig 2 place M0 against the development comparators that were actually run. Urdu-only BM25 (no Roman path) scored 0.5897. Headline MiniLM scored 0.4487. The truncated full-article dense index scored 0.2564, worse than headlines, consistent with truncation at 128 tokens. Chunk ANN scored

Table 1. Official M0 results. ExactSource Hit@5 and human Success@5 answer different questions. Intervals are Clopper–Pearson 95% intervals computed from the reported counts. Do not average rows. Do not treat 87.18% as unseen or as human usefulness.

Evaluation	Dataset	Metric	Result
Development/ validation known-item	Phase 2, $n = 78$	ExactSource Hit@5	68/78 (87.18%; 77.68–93.68%)
New known-item	K001–K040	ExactSource Hit@5	27/40 (67.50%; 50.87–81.43%)
New naturalistic (human)	U001–U040	Success@5	23/40 (57.50%; 40.89–72.96%)

0.2821. Those dense numbers are development evidence that a full MiniLM index was not a substitute for a Roman lexical path. They are not Phase 12 results.

Table 2. Development/validation comparators on the same $n = 78$ known-item pool. Official M0 is the script-aware row. Oracle unions were not deployed.

System	Hit@5	nDCG@5	MRR
Truncated full MiniLM (Chroma)	0.2564	0.2203	0.2103
Chunk ANN (96/32)	0.2821	0.2362	0.2309
Headline MiniLM	0.4487	0.4009	0.3885
Urdu BM25, no Roman path	0.5897	0.5509	0.5425
Script-aware M0	0.8718	0.8107	0.797
Headline + M0 oracle (not deployed)	0.9103	0.8703	0.8615

Fig 2. ExactSource Hit@5 on the Phase 2 development/validation pool ($n = 78$). Values are the frozen development comparators in Table 2. Script-aware M0 is the official system. This figure is not a Phase 12 result.

On the 46 Urdu-script queries in that pool, Urdu BM25 Hit@5 was 0.913; sending those queries to the same Urdu index left that number unchanged. On the 23 Roman queries, Method A was 0/23 and headline MiniLM was 1/23. Method D recovered 22/23 (development 13/13, internal validation 9/10). The remaining miss, QTRN_031, was rank 9 with 10 overlapping tokens against the romanized source: Hit@10 succeeded, so the residual was ranking competition rather than a remaining script mismatch. Method B remained 0/13 on development Roman queries: 22 of 23 Roman strings were still mostly Latin after dictionary lookup. Method C tied Method D on development Hit@5 (13/13) but lost on nDCG@5 (0.8004 versus 0.9331) and would have scored 6/10 on internal validation in a post-hoc diagnostic that was not used for selection. Mixed queries ($n = 9$) scored Hit@5 = 0.4444 on the Urdu path; a mixed-path union did not raise that figure.

A later rebuild (Phase 6) reproduced 68/78 and listed the ten residual misses: four URDU, one ROMAN (QTRN_031), and five MIXED. Four of those ten sat in ranks 6–10. Mixed-script items in this pool were generated with an English template suffix, so those misses are not a census of naturally mixed user queries. A qualitative primary label was assigned to each of those ten residuals (query ambiguity, wrong index room, entity collision, topical neighbour, or known-item ambiguity among near-duplicate wires). Those labels describe that freeze-pool remainder. They are not a rate, and they are not the Phase 12 diagnosis: sealed Roman misses are mostly sources that never enter the Top-50.

The 87.18% result is genuine for title-derived known-item search on this pool, including Roman queries that resemble Method D's own romanization. It is not a forecast of chat-style Roman performance.

Phase 11 ablation

Table 3 reports the query-side expansions. M0 through M4 all scored 68/78. Train-Roman Hit@5 stayed 61/64 (95.31%). M1–M4 nDCG@5 and MRR on that diagnostic were slightly below M0 (0.8940 and 0.8781 versus 0.8960 and 0.8807). No candidate beat 68/78. M1 passed the gate as the simplest non-destructive candidate and was not installed as the official system.

Table 3. Phase 11 query-side Roman expansions. The official system remains M0. Train Roman $n = 64$ is a selection diagnostic, not an unseen test.

Model	$n = 78$ **Hit@5**	**Train Roman Hit@5**	**Train nDCG@5**	**Train MRR**
M0 (official)	68/78	0.9531	0.8960	0.8807
M1	68/78	0.9531	0.8940	0.8781
M2	68/78	0.9531	0.8940	0.8781
M3	68/78	0.9531	0.8940	0.8781
M4	68/78	0.9531	0.8940	0.8781

New known-item (K001–K040)

Preflight passed: corpus and dictionary hashes matched the freeze, BM25 parameters were unchanged, and M1–M4 were off. Every query returned 50 hits. Detector counts were 28 URDU and 12 ROMAN, matching the retrieval-path counts.

Table 4 reports ExactSource Hit at four cutoffs. The primary metric is Hit@5 = 27/40. Hit@1 is 20/40. Two further sources appear between ranks 6 and 50 (Hit@10 = 28/40, Hit@50 = 30/40). Ten of the 13 Hit@5 misses are absent from the entire Top-50; three are in the list but below rank 5 (K002 rank 6, K010 rank 49, K031 rank 17). Per-query source ranks are in S1 Table.

Table 4. Phase 12 K ExactSource results, frozen M0. Primary metric is Hit@5. This evaluation does not replace 68/78 and is not human Success@5.

Metric	**Result**
ExactSource Hit@1	20/40 (50.00%)
ExactSource Hit@5	27/40 (67.50%; 50.87–81.43%)
ExactSource Hit@10	28/40 (70.00%)
ExactSource Hit@50	30/40 (75.00%)

The detector split was not used for tuning. Urdu-script K titles scored 26/28 ExactSource Hit@5 (92.86%; 76.50–99.12%). Ordinary Roman titles scored 1/12 (8.33%; 0.21–38.48%). Fig 3 shows that split beside the naturalistic script split. Fig 4 separates Hit@5 from “source present but below rank 5” and “source absent from the Top-50.” Both Urdu misses (K002 rank 6, K010 rank 49) are still in the list. Ten of eleven Roman misses never enter the Top-50; the remaining Roman miss (K031) is rank 17. The drop from 87.18% to 67.50% is concentrated on ordinary Roman title queries, and on matching rather than near-miss ranking, not on native-script Urdu BM25.

Fig 3. Descriptive script splits of frozen M0, not used for tuning. Left: ExactSource Hit@5 on K001–K040. Right: human Success@5 on U001–U040. Mixed $n = 4$ is descriptive only. The two panels are different metrics and must not be averaged.

Fig 4. Where sealed known-item misses sit. Counts from the K001–K040 per-query source ranks. Urdu misses remain inside the Top-50. Most Roman misses never appear in the retrieved 50.

Naturalistic human evaluation (U001–U040)

Success@5 was 23/40. Conservative P@5 was 0.2050 (variable-denominator P@5 was identical because every query had at least five hits). nDCG@5 was 0.6460. MRR was 0.4542. Among 200 labeled documents the counts were A 41, B 26, C 53, D 80, E 0 (Table 5; Fig 5). Forty-one fully answering documents across 40 queries is consistent with conservative P@5 = 0.2050: many Top-5 lists contain at most one A.

Table 5. Phase 12 U human metrics, frozen M0 Top-5. Success@5 is the usefulness headline. nDCG@5 includes gain 1 for topical C and is not a usefulness claim.

Metric	Result
Success@5 (any A or B)	23/40 (57.50%; 40.89–72.96%)
Conservative P@5 (A-count/5)	0.2050
nDCG@5 (A=3, B=2, C=1)	0.6460
MRR (first A or B)	0.4542
Label counts (200 documents)	A 41, B 26, C 53, D 80, E 0
All-D queries	12/40

Fig 5. Label mass on the 200 judged U Top-5 documents. A = relevant, B = partially relevant, C = topically related, D = not relevant, E = ambiguous. E did not occur. D is the modal label.

The gap between Success@5 and conservative P@5 is the shape of many successes: one useful document rather than a Top-5 packed with full answers. nDCG@5 overstates usefulness for the same reason the gain table is generous to C.

Fig 3 and Table 6 show descriptive slices. Urdu-script queries succeeded in 17/18 cases (94.44%; 72.71–99.86%). Roman queries succeeded in 6/18 (33.33%; 13.34–59.01%). Mixed queries were 0/4 (0–60.24%). Mixed $n = 4$ is too small for a population rate; it is reported because all four failed. These slices describe this sealed sample. They are not a licence to retune Method D on U failures.

Table 6. Descriptive U Success@5 slices. Script uses the detector label at retrieval time. Need type and length follow the sealed query file.

Slice	Successes	n	Rate
URDU	17	18	0.9444
ROMAN	6	18	0.3333
MIXED	0	4	0.0000
Factoid	9	14	0.6429
Explanatory	9	14	0.6429
Named entity	5	12	0.4167
Short ($\leq$ 5 tokens)	9	12	0.7500
Medium (6–12)	6	16	0.3750
Long (13+)	8	12	0.6667
Temporal	3	4	0.7500
Non-temporal	20	36	0.5556

Complete Success@5 failures were U004, U006, U008, U010, U014, U016, U018, U020, U026, U028, U032, U034, U035, U037, U038, U039, and U040. Twelve of those lists were all D, including several English-loan or service queries (`psl points table`, train-ticket price, Netflix charges, 5G, dollar rate) and several chat-style Roman strings. Per-query labels are in S2 Table.

Diagnostic H001–H040

Phase 10C human Success@5 on H001–H040 was 25/40 (62.50%; 45.80–77.27%), with conservative P@5 = 0.1250 and 10/40 all-D lists. Urdu-script traps succeeded in 14/20 cases and Roman traps in 11/20;

eight of the ten all-D lists were Roman. H036 returned one document, labeled D. The queries are historical trap strings, not the Phase 12 naturalistic sample. The number is recorded so it is not silently dropped; it is not the official unseen usefulness result.

Discussion

The three headline percentages look like a system falling apart. They are three measurements.

On the freeze pool, M0 often recovers a known article from a shortened headline, including Roman queries built as `title_roman`. Method D was selected on that construction. High Hit@5 is what one should expect when the query is a title fragment and, for Roman, when the spelling family matches the index. RQ1 is answered in that setting: 68 of 78 sources reached the Top-5, well above Urdu-only BM25 on the same pool, because the Roman path repaired a script mismatch that Method A could not (0/23). That comparison is the available evidence that a single Urdu BM25 index is insufficient for this Roman construction. A no-Roman-path ablation was not repeated on sealed K or U; those sets were not used to retune the freeze.

K keeps the known-item question and changes the sample. Roman K queries were written as ordinary Roman Urdu of the headline, not as character-table `title_roman`. Urdu K stayed high (26/28), and both Urdu misses were still in the Top-50. Roman K did not (1/12), and ten of those eleven misses never entered the Top-50. That pattern is a matching failure, not a near-miss ranking problem. A reranker of the existing Top-50 cannot recover a source that is absent from the list. The 19.7 point drop from 87.18% to 67.50% is therefore largely a Roman query-form shift, plus ordinary binomial variation on $n = 40$. It is not evidence that Urdu BM25 broke. RQ2 is answered in the negative if "transfer" means the 87% figure itself; the honest unseen known-item figure on this sealed K set is 27/40.

U asks a different question. There is no single right article. Factoids, explanations, named entities, underspecified strings, and chat Roman are harder than "find this headline." Success@5 only requires one A or B, and still lands at 23/40. Conservative P@5 of 0.205, with 41 A labels in 200 documents, shows that the Top-5 is rarely full of complete answers; D is the most common label. Named-entity queries in this sample were weaker (5/12) than factoid or explanatory queries (each 9/14). Urdu-script needs were usually met (17/18). Roman needs often were not (6/18). RQ3 is therefore a usefulness rate of 23/40 overall, and 17/18 when the user types Urdu script, on this $n = 40$ set with official Annotator-1 labels. Those slice rates have wide intervals and are not population estimates. Independent second-annotator Success@5 is a reliability statistic (S2 File) and does not replace 23/40. A2 does not show that A1 is unbiased: Annotator 1 still wrote the queries.

Query mix will move any headline number. A Roman-heavy sample will look worse than an Urdu-only test even if the Urdu component is unchanged. Averaging 87.18% with 57.50% answers no scientific question: it mixes development with new test, known-item with graded usefulness, and `title_roman` with chat Roman.

RQ4 is straightforward. Allowed query-side expansions did not move 68/78. A small spelling table is not a demonstrated fix for the Roman gap on K and U, and those sets were not used to invent a larger table.

The pattern is consistent with how Method D was built. Document romanization plus a 198-key reverse dictionary can match pipeline-romanized titles. It does not enumerate WhatsApp-style spelling, English loanwords that users type in Latin while the article uses Urdu, or mixed-script strings that the freeze sends to the Urdu index. Method B already showed that dictionary lookup on the query is too sparse for naive romanizations. The freeze kept the document-side index. That decision was justified for `title_roman`. It was not a solution to open-vocabulary Roman IR.

Relative to prior Urdu resources, these numbers are not a CURE or MS MARCO leaderboard entry [5,6]. The protocol is different, the collection is news, and the Roman path is the scientific point. Relative to English routers that choose sparse versus dense retrieval [11], M0 only chooses which script index to open. That is a smaller engineering rule, implemented by Unicode counts, not a learned retrieval policy.

What remains inspectable is native-script Urdu BM25 on this news collection: it recovers title-like known items and, in this U sample, usually puts something useful in the Top-5. Script-aware index selection repairs the development `title_roman` mismatch that a single Urdu BM25 index does not (Method

A 0/23). It does not repair ordinary or chat-style Roman Urdu on K and U. The study is still useful as a frozen baseline and as an evaluation protocol that keeps those facts separate. Future work may investigate first-stage Roman matching, mixed-script paths, or denser indexes; those are not claims about M0.

Limitations

K and U each have $n = 40$. Roman K has $n = 12$. Mixed U has $n = 4$. The intervals in Table 1 are wide. Point estimates should not be read as a precise "58% in the wild," and script slices are descriptive of this sample only.

Official U labels (A1) were assigned by the first author, who had also written the sealed U queries under the Phase 12 protocol. An independent second annotation (A2) was later conducted by Areena Rahman on the same 200 judgments as a reliability check (S2 File); five-way Cohen's $\kappa = 0.5490$ and binary $\kappa = 0.6816$. A2 was not used to replace or recompute 23/40, was not used to tune M0, and does not remove the dual-role bias risk. Preferring B over A reduces over-claiming of full answers; it does not remove subjectivity. Labels were assigned after retrieval, from headline and snippet only, and without searching for a better document.

Development Roman queries are `title_roman`. K Roman queries are ordinary Roman titles. U Roman queries are chat-style. Those are three related but distinct input families. Success on the first does not contradict failure on the other two. Method D was selected on $n = 13$ development Roman queries.

The nine MIXED items in the $n = 78$ pool were generated with an English suffix. Phase 12 mixed queries ($n = 4$) are too few to estimate a rate. Both facts limit what can be said about mixed-script search.

The corpus is third-party Urdu news only. Cleaning was a null drop. Near-duplicate wires were not collapsed. Results should not be read as general Urdu IR, other domains, other languages, or operational search logs. M0 does not rewrite queries, does not use dates, and does not generate answers. Temporal "today" items were judged as archive type-of-fact.

nDCG@5 with C-gain = 1 can look high when Success@5 fails. H001–H040 are diagnostic only. K, U, and H001–H040 are burned if M0 is changed.

Sealed K and U do not include a repeated Urdu-only BM25 or multilingual dense, hybrid, or reranking run. Development Table 2 already compares Urdu-only BM25 and MiniLM variants on $n = 78$. A CPU `multilingual-e5-small` index was not built. Those absences bound claims about this index-selection rule versus modern neural IR; they do not change the frozen M0 numbers.

The methodological novelty is modest: deterministic script detection plus BM25. That is a limitation of the contribution, not a defect in the freeze.

The git branch `research/post-phase12` contains later exploratory development work conducted after the frozen publication evaluation. Those experiments were not used to tune or replace the official M0, K, U, or A1 results reported in this manuscript and are not treated as official test results. They are not combined with 68/78, 27/40, or 23/40.

We do not claim 80% unseen usefulness, 87.18% real-world accuracy, or state-of-the-art ranking against CURE or Urdu MS MARCO.

Conclusion

Deterministic script-aware BM25 is a simple, hashed baseline for Urdu news search: Unicode counts select a native-script index or a Method D romanized-document index. On the development/validation known-item set the frozen system recovered the source article in the Top-5 for 68 of 78 title-derived queries. On a newly sealed known-item set it did so for 27 of 40. On a separate naturalistic set, official Annotator-1 labels found at least one useful document in the Top-5 for 23 of 40 queries. Native-script title-like search on this collection often recovered the designated article. Ordinary Roman Urdu did not. The study does not solve Urdu or Roman Urdu retrieval. It provides reproducible evidence, under a freeze, of where this lexical pipeline works and where later first-stage matching work is needed.

Supporting information

S1 Table. Per-query ExactSource ranks for K001–K040. Query text, detector label, retrieval path, source document identifier, source rank, and Hit@5 for each sealed known-item query. File: `S1_table.csv`.

S2 Table. Per-query official human labels for U001–U040 (Annotator 1). Raw query text, five Top-5 labels (A–E), Success@5, rank of first A/B, conservative P@5, nDCG@5, and MRR. Official Success@5 is 23/40. File: `S2_table.csv`.

S3 Table. Phase 5 development Roman method comparison. Methods A–D on DEV Roman queries ($n = 13$) with Hit@5, Hit@10, nDCG@5, MRR, and mean latency, plus internal-validation confirmation of Method D ($n = 10$). File: `S3_table.csv`.

S4 Table. Annotator 1 versus Annotator 2 per-query labels for U001–U040. Five A1 labels, five A2 labels, Success@5 for each annotator, and document-level agreement counts. A2 Success@5 is reliability only and does not replace 23/40. File: `S4_table.csv`.

S1 File. Freeze manifest. Corpus SHA-256, document count, BM25 parameters ($k_1 = 1.5$, $b = 0.75$), index statistics, dictionary size, and script-to-index table. Historical freeze JSON; the `test_set` field still names H001–H040. Official unseen evaluations are K001–K040 and U001–U040. File: `S1_file.json`.

S2 File. Independent second annotation (reliability analysis). A2 was conducted by Areena Rahman. Five-way and binary agreement, Cohen's κ (0.5490 and 0.6816), confusion matrices, and the statement that A2 Success@5 = 26/40 does not replace official A1 Success@5 = 23/40. No article text. File: `S2_file.md`.

S3 File. Dataset provenance and reconstruction notes. Third-party Kaggle/Mendeley source, 111,861→111,860 truncated-row drop, frozen SHA-256, and instructions to verify a reconstructed corpus. Does not contain `clean_articles.csv` or any full article collection. File: `S3_file.md`.

S1 Text. U annotation protocol (Annotator 1). Label definitions A–E, temporal type-of-fact rule, named-entity rule, and metric formulae for Success@5, P@5, nDCG@5, and MRR. File: `S1_text.md`.

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